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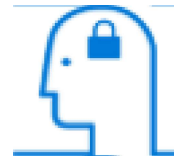
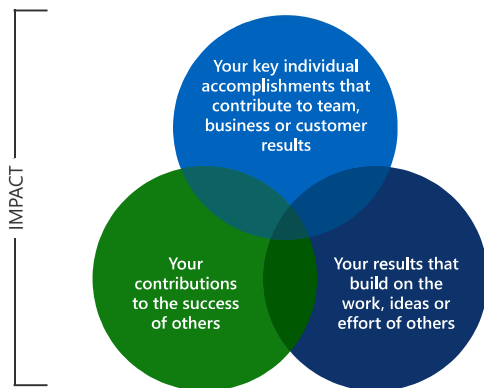
DEVESH KUMAR

SOFTWARE ENGINEER II | Reports to: Suma Kollipara

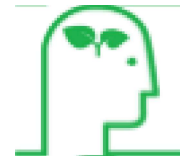
Connect May 2024

Reflection Period: Oct 24, 2023 - May 11, 2024

Connects provide a regular touchpoint where you and your manager can come together with a growth mindset to reflect on impact and discuss areas for improvement and future development.



Fixed Mindset leads to a desire to look smart and therefore a tendency to avoid challenges, give up easily, see failures as fruitless or worse, ignore useful negative feedback and feel threatened by the success of others.



Growth Mindset leads to a desire to learn and therefore a tendency to embrace challenges, persist in the face of setback, see failures as essential to mastery, learn from criticism and find lessons and inspiration in the success of others.

Reflect on the past

One aspect of successful impact is when you consistently deliver against all and likely exceed some expectations for your core priorities. Capture what you've accomplished and learned since your last Connect.

Summarize your impact

Core priorities

What impact did you have for each of your core priorities?

Coding work:

- **Removing Azure digital Twin Nov 23- dec 23:**

- Modified the ingestion architecture to remove Azure Digital Twin. This greatly helped in scaling of ingestion flow to handle larger ingestion rate.
- This also helped in reducing the cost as Azure digital twin was most cost heavy resource that we were using.
- Modified all the different flows to use direct ingestion in ADX. These ingestion flows include:
 - Node creation
 - Relationship creation
 - OPCUa data creation
- Modified the groupby clause to process multiple twins in a single call to ADX. This greatly helped in increasing speed of ingestion of relationship as the group by clause was a bottleneck.
- Implemented Bicep changes for the update entity flow to enable direct ingestion in Azure Data Explorer
- Optimized the ingestion host and Function App configuration settings for optimal performance
- All the above optimizations and architectural changes enabled us to reduce the ingestion time from 18 minutes to 11 minutes for 330k nodes. Additionally, we improved the relationship ingestion time from 39 minutes to 18 minutes
- **Vectorization in ingestion flow** *Jan 24*
 - Ingestion flow
 - Revised the ingestion flow to perform vectorization of selected entities concurrently with ingestion, a task previously handled by a background job of copilot service.
 - Leveraged Redis to store metadata for schema and relationship vectorization, thereby increasing the throughput for vectorization.
 - Implemented a retry mechanism for Copilot calls to manage throttling issues.
 - DMM Service
 - Introduced an API to modify the 'isProperNoun' fields for columns in both inbuilt and custom entities.
 - This field allows customers to vectorize certain columns of their data, thereby enhancing the performance of Copilot
 - Inbuilt Job
 - Configured the inbuilt job to generate the necessary mappings and tables for vectorization.
 - Implemented the necessary Bicep changes to enable Function App to access Azure OpenAI through Managed Identity.
 - Decoupled the vectorization flow entirely to prevent interference with the regular ingestion flow. To achieve this, we introduced a dedicated Event Hub solely for vectorization.
- **Logging improvement** *Jan 24*
 - Improved logging within the ingestion flow by eliminating excessive and redundant logs
 - Introduced logs within the ingestion flow for critical pathways.
- **Migration to dotnet8** *Feb 24*
 - Migrated the codebase including SKG, copilot and ingestion from dotnet 7 to dotnet 8. This would enable for enhanced security and compliance.
 - Did the relevant bicep changes too so that the function app and AKS pods are also updated.

POC:

- **Copilot pipeline changes** *April 24*
 - Did changes to copilot python script.
 - Tried multiple approaches with include:
 - Introduced keywords in instruction for getting better results in similarity search
 - Breaking down Instruction into sub instruction and then doing similarity search
 - Introduced graph query in python script. Through graph query we get better copilot result with less token as it would eliminate the pseudocode layer of copilot.
- **IOT devices POC** *March 24*

- Customers with their own IoT devices may wish to integrate their data with DMM. To facilitate this, I did an POC to establish a data pipeline for their Azure IoT devices.
- I configured an MQTT broker on my machine to emit OPCUA data, and subsequently adjusted the DMM service to process this data
- **ADLS and ADX cost Comparision**
 - Authored a Markdown document comparing the cost of storing approximately 50 million rows of data in Azure Data Lake Storage (ADLS) and Azure Data Explorer (ADX)
 - This analysis led to the elimination of external tables in ADX that were storing data in ADLS. This will

Bug Fixes:

- **Parquet Reader Bug**
 - Encountered a bug in parquet reader flow where the code was not able to process the parquet file as the cursor for the reader was incorrectly positioned at the end of the file, causing an EOFException. Fixed that bug and merged it to release branch.
- **Adx having empty values**
 - ADX was having empty values in model ID section for deleted relationship.

ThreatModel:

- **Completed Threat modeling course**
 - In last semester I completed threat modeling course and passes the table top exam following that.
 - After completing this course, I could review architecture of different teams under MCI team for any threats. This would not only enforce a better security practice across MCI team but also help me learn about different architecture different teams are taking in MCI.
- **Industry AI Team**
 - Did threat model review for industry AI team for their product named DTT/ DMF (Data Movement Framework)

Diversity & Inclusion (D&I) Core Priority

What impact did your actions have in contributing to a more diverse and inclusive Microsoft?

- Participated team collaboration across MCI in team building activities
- Mentored new hires during their initial onboarding process, providing guidance and support in setting up and understanding the codebase.
- Participated in the hiring process for the sustainability team, conducting interviews and evaluating candidates based on their technical skills and cultural fit.



Manager comments

Devesh is extremely high velocity contributor in SKG team who always tries to become a better developer each single day. His passion for picking up challenges with ADT removal work, copilot V2 enhancement experimentation and many others and tackling them systematically and delivering impact with consistency. He is highly accountable and no matter

what delivers the assigned task on time with quality. He always makes sure he picks up different kinds of work and that helped him to become part of Threat modelling team at MCI, switch streams for accepting copilot work and delivering within a very short time, and also become go to person for lot of data mapping related queries.

Reflect on a challenge or setback

In addition to your accomplishments relative to your core priorities, you must demonstrate a growth mindset to deliver successful impact. Share how you demonstrated a growth mindset by taking personal accountability for setbacks, asking for feedback, and applying learnings to have greater impact.

Consider when you could have done something differently. How will you apply what you learned to make an even greater impact?

- While configuring jobs for the vectorization table, I initially overlooked the scenario where users enable the copilot after deployment. To rectify this, I adjusted the job to execute whenever the copilot is toggled.
- During the vectorization process, we encountered throttling issues with the copilot, particularly when dealing with large initial data sets. For overcoming this we added extra retry logic for Azure OpenAI calls.
- For copilot python script took a lot of time as there were many iterations.



Manager comments

I acknowledge with the reflection of where there could have been a better job especially in the vectorization for copilot data.

Plan for the future

Think about opportunities for impact in the upcoming Connect period. What are your 3-5 core priorities?

Review your core priorities

Core priorities

1. Better Design Patterns

Focus areas for the upcoming period to drive business impact:

- I will try to incorporate the best design patterns in the DMM codebase.
- Study other codebases on how they implement the similar work and try to bring the best coding practices in DMM .

- Enrich myself on latest design patterns through online courses.
- Review already implemented and ongoing design of DMM features and bring improvements in it.

Success measures:

- Better overall design of the DMM product.
- Easy maintainability and debugging of the DMM product.

2. Security features and Threat modeling

Focus areas for the upcoming period to drive business impact:

- Get up to speed on security work and features developing on security wave 1 exercise. Try to integrate the knowledge gained in threat model in security wave 1 and wave 2 features.
- Threat modeling:
 - Focus on conducting at least 2 reviews in month for threat model
 - This will enable me to dive into various architecture diagram of different teams. Hopefully i could bring some of the best practices back to our team too
 - Overall enhancement of security across MCI team.

Success measures:

1. Better security in architecture level
2. More understanding of different architecture design
3. Understanding about new technologies and pattern that others are using

3. Copilot changes and scale motion

Focus areas for the upcoming period to drive business impact:

- In last semester we did POC to improve the latency and the accuracy of the copilot. We did various POC on how to breakdown instructions and use keywords.
- Next semester I will focus on implementing them in our main code as currently we are trying this on the python script only.
- The next semester is about scale motion for DMM. I we are focusing on finetuning our codebase and documentation improvements
I will also focus on adhering to release timelines and take up DRI work.

Success measures:

If the accuracy and latency is improved in actual customer usage
If more users are using copilot.
Faster DRI resolution

4. GSI and customer engagement

Focus areas for the upcoming period to drive business impact:

Take part in Customer engagements. Assist atleast one customer for deploying DMM end to end. Assist them in data mapping and Copilot setup.

Success measures:

Better overall customer adoption and satisfaction.

Diversity & Inclusion (D&I) Core Priority

To realize our mission to empower every person and organization on the planet to achieve more, each and every one of us must help create a diverse and inclusive culture where everyone can bring their full and authentic self, in alignment with our values. Consider how you will **deliver successful impact by consistently demonstrating Microsoft's cultural expectations and values**. Visit the [Diversity & Inclusion \(D&I\) Core Priority page](#) for guidance and examples of impact.

Select and commit to at least one focus area from the options below, or create one of your own.

- Seek diverse perspectives and incorporate feedback, apply insights from a variety of sources and people
- Mentor and/or support a person who may benefit

How will you define success as you activate your D&I Core Priority?

- The perspective and feedback from others would be a good measure for success.



Manager comments

Completely aligned with the priorities for the upcoming period and looking forward for Devesh's impact in our next big milestones for DMM

Identify your opportunities for growth

We are all expected to embrace a growth mindset and focus on continuous learning to deliver successful impact.

What 1-2 actions will you take to focus on learning, growth, and development over the next period?

- I will try to get deeper into new methods that are coming up for copilot.
- Understanding more in copilot codebase.

What support do you need to fulfill these actions? Fill in any that apply.

Opportunities to (optional):

No Response

Perspective or coaching on (optional):

No Response

Learning on (optional):

No Response



Manager comments

Absolutely, would love to see you becoming an SME for Generative AI offering and our copilot solutions.

Conversation starters

Choose any of these optional prompts to continue great conversations with your manager. Write a brief response (1-15 words).

Of my strengths, the ones I'd like to apply more in my work are:

I would like to work on my coding and designing skills.

In my work, I feel motivated or energized by:

When I get new and challenging work where I can learn new technologies.

In my work, I lose motivation or energy from:

When there are too many group calls with large no of audience.

It can be challenging to prioritize important topics for 1:1 conversations. Plan ahead and flag any of the following topics you'd like to discuss with your manager in upcoming conversations.

Select any conversation topics below, or add your own:

Career: Discuss ways to grow my career to align with my strengths, interest & aspirations.

Recognition: Discuss my [recognition preferences](#) and highlight my contributions.

Opportunities: Increase my responsibilities, such as new core priorities, stretch assignments, promotion or becoming a people manager.

Check out aka.ms/career for resources and guides on developing your career.



Manager comments

Acknowledged and in-alignment to work with you and support you as per your preferences and carving out your next career milestones.